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TEST REPORT	
Report Reference No.:	2NL202133001
Date of issue.....:	20/02/2024
No. of Pages	11
Testing Laboratory Name	OTL Testing Labs Pvt. Ltd., Unit 2
Address	56, Industrial Area, Phase-I, Chennai-600 001, India
Manufacturer's Name	XYZ Private Limited
Address.....	ABC
Applicant's Name	XYZ Private Limited
Address	ABC
Test Report Form No:	CTL_PAR_TRF_V10.000
Master TRF Date:	2018.03
Test Report Form(s) Originator:	OTL Testing Labs Pvt. Ltd., Unit 2, New Delhi
Test item description	Electrocardiograph
Trade Mark	NIL
Model/Type reference	HC102
Ratings	100-240V~ 50/60Hz 130VA Class I
Serial No	OC0A2:22260
Applicable Standard Specification	IS 13450 (Part 1) : 2018/ IEC 60601-1 : 2012 As per customer request: • Power input • Earth impedance • Leakage currents • Di electric strength Test • Interruption of power supply • Mechanical test-Drop • Mechanical test- Impact

For and Behalf of OTL Testing Labs Ltd.

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Designation: Director Technical
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Summary of testing:		
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1.	Power input test as per Cl. 4.11 of IS 13450 (Part 1) : 2018/ IEC 60601-1 : 2012	04
2.	Impedance and current-carrying capability (earth impedance) as per Cl. 8.6.4 of IS 13450 (Part 1) : 2018/ IEC 60601-1 : 2012	04
3.	LEAKAGE CURRENTS and PATIENT AUXILIARY CURRENTS as per Cl. 8.7 of IS 13450 (Part 1) : 2018/ IEC 60601-1 : 2012	04
4.	Dielectric strength test as per Cl. 8.8.3 of IS 13450 (Part 1) : 2018/ IEC 60601-1 : 2012	04
5.	Interruption of power supply as per Cl. 11.7 of IS 13450 (Part 1) : 2018/ IEC 60601-1 : 2012	05
6.	Mechanical strength Test- Drop Test Cl. 15.3.4.2 of IS 13450 (Part 1) : 2018/ IEC 60601-1 : 2012	05
7.	Mechanical strength Test- Impact Test Cl. 15.3.3 of IS 13450 (Part 1) : 2018/ IEC 60601-1 : 2012	05

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Possible test case verdicts:

- test case does not apply to the test object : N/A
- test object does meet the requirement : P (Pass)
- test object does not meet the requirement : F (Fail)

Testing :

Date of receipt of test item : 30-05-2023

Date (s) of performance of tests..... : 01-06-2023 to 12-06-2023

Condition of Samples.....: Working condition

Laboratory conditions:

Ambient Temperature.....: (25 ± 10)°C

Ambient Humidity.....: (60 ± 15)% RH

General remarks:

1. The test results presented in this report relate only to the object tested.
2. This report shall not be reproduced, except in full, without the written approval of the issuing testing laboratory.
3. All the Tests has been performed on the submitted sample.

Throughout this report a ☐ comma / ☒ point is used as the decimal separator.

General product information:

XXXXXXX HE 1200 is digital 12 channel physiological function measurement equipment which records the heart waveform during electro-biological movement. It provides sufficient information on the analysis of arrhythmia and cardiovascular disease.

Type CF applied parts: patient Leads

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Discipline: Electronics

Group: Medical Electrical Equipment

Sl. No:	Requirement +Test	Measured Value/ observations	Verdict
1.	Power input test as per Cl. 4.11 of IS 13450 (Part 1) : 2018/ IEC 60601-1 : 2012 The steady-state measured input of the ME EQUIPMENT OR ME SYSTEM at RATED voltage and at operating settings indicated in the instructions for use shall not exceed the marked rating by more than 10 %	Refer Table 4.11 below	P
2.	Impedance and current-carrying capability (earth impedance) as per Cl. 8.6.4 of IS 13450 (Part 1) : 2018/ IEC 60601-1 : 2012 the impedance between the protective earth pin in the MAINS PLUG of any DETACHABLE POWER SUPPLY CORD supplied or specified by the MANUFACTURER, when attached to the ME EQUIPMENT, and any part of the ME EQUIPMENT that is PROTECTIVELY EARTHED shall not exceed 200 mΩ .	Measured Impedance:80.3mΩ	P
3.	LEAKAGE CURRENTS and PATIENT AUXILIARY CURRENTS as per Cl. 8.7 of IS 13450 (Part 1) : 2018/ IEC 60601-1 : 2012	Refer Table 8.7 below	P
4.	Dielectric strength test as per Cl. 8.8.3 of IS 13450 (Part 1) : 2018/ IEC 60601-1 : 2012 Test voltage: 3000V for 1 min During the test, breakdown constitutes a failure.	No breakdown occurred	P

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5.	Interruption of power supply as per Cl. 11.7 of IS 13450 (Part 1) : 2018/ IEC 60601-1 : 2012 After the restart the ME Equipment should maintain basic safety and essential performance	In compliance basic safety and essential performance maintained . Essential performance checked at 1mV Normal sinus rhythm signal after the test.	P
6.	Mechanical strength Test- Drop Test (portable equipment) as per Cl. 15.3.4.2 of IS 13450 (Part 1) : 2018/ IEC 60601-1 : 2012 Test Condition: from height 5 cm After the test, any damage sustained that results in a unacceptable RISK constitutes a failure.	In compliance, No damage that constitutes an unacceptable RISK or loss of essential performance. Essential performance checked at 1mV Normal sinus rhythm signal after the test.	P
7.	Mechanical strength Test- Impact Test as per Cl. 15.3.3 of IS 13450 (Part 1) : 2018/ IEC 60601-1 : 2012 Test Condition: Steel ball (50 mm in dia., 500 g $\pm$ 25 g) falling from a 1.3 m After the test, any damage sustained that results in a unacceptable RISK constitutes a failure.	In compliance, No damage that constitutes an unacceptable RISK. Essential performance checked at 1mV Normal sinus rhythm signal after the test.	P

Attachment 1: Tables

4.11	TABLE: Power Input				P
Operating Conditions / Ratings	Voltage (V)	Frequency (Hz)	Current (A)	Power (VA)	Power factor (cos ϕ)
RATED power 130VA	100	50	0.225	22.50	0.45
	100	60	0.225	22.50	0.45
	240	50	0.133	31.92	0.41
	240	60	0.133	31.92	0.41
Supplementary Information:					

8.6.4	TABLE: Impedance and current-carrying capability of PROTECTIVE EARTH CONNECTIONS			
Type of ME EQUIPMENT & impedance measured between parts	Test current (A) /Duration (s)	Voltage drop measured between parts (V)	Maximum calculated impedance (m Ω)	Maximum allowable impedance (m Ω)
ME EQUIPMENT with a DETACHABLE POWER SUPPLY CORD,	25A/10s	2.4	80.3	200

Supplementary information:

ME EQUIPMENT with a DETACHABLE POWER SUPPLY CORD, impedance between the protective earth pin in the MAINS PLUG and a PROTECTIVELY EARTHED part - Limit 200 m Ω

8.7	TABLE: leakage current			P
Type of leakage current and test condition (including single faults)	Supply voltage (V)	Supply frequency (Hz)	Measured max. value	Remarks
Fig. 13 - Earth Leakage (ER)	—	—	mA	Maximum allowed values: 5 mA NC;
NC S5=1;S7=1;	240	50	0.237	In compliance
NC S5=1;S7=0;	240	50	0.372	
NC S5=0; S7=1	240	50	0.238	
NC S5=0; S7=0	240	50	0.375	
Fig. 14 - Touch Current (TC)	—	—	μA	Maximum allowed values: 100 μA NC; 500 μA SFC
NC S5=1;S7=1;	240	50	16.6	In compliance
NC S5=1;S7=0;	240	50	22.9	
NC S5=0; S7=1	240	50	17.1	
NC S5=0; S7=0	240	50	24.7	
Fig. 15 - Patient Leakage Current (P)	—	—	μA	Maximum allowed values: Type B or BF AP: 10 μA NC; 50 μA 100 μA NC; 500 μA Type CF AP: 10 μA NC; 50 μA
NC S5=1;S7=1;	240	50	1.27	In compliance, performed on patient leads, Type CF applied parts
NC S5=1;S7=0;	240	50	17.3	
NC S5=0; S7=1	240	50	2.4	
NC S5=0; S7=0	240	50	18.4	
Fig. 16 - Patient leakage current with mains on the F-type applied parts (PM)	—	—	μA	Maximum allowed values: Type B: N/A Type BF AP: 5000 μA Type CF AP: 50 μA
NC S5=1;S7=1;	240	50	13.2	In compliance, performed on patient leads, Type CF applied parts
NC S5=1;S7=0;	240	50	17.3	
NC S5=0; S7=1	240	50	13.5	
NC S5=0; S7=0	240	50	17.6	
Fig. 17 - Patient leakage current with external voltage on Signal Input/output part (SIP/SOP)	—	—	μA	Maximum allowed values: Type B or BF AP: 10 μA NC; 100 μA NC Type CF AP: 10 μA NC; 50 μA SFC
NC S5=1;S7=1;	240	50	0.7	In compliance
NC S5=1;S7=0;	240	50	0.78	
NC S5=0; S7=1	240	50	0.81	

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NC S5=0; S7=0	240	50	0.91	
Fig. 18 - Patient leakage current with external voltage on metal Accessible Part that is not Protectively Earthed	—	—	—	Maximum allowed values: Type B or BF AP: 500 μ A Type CF: N/A
				N/A, all metal accessible parts are protectively earthed
Fig. 19 – Patient Auxiliary Current	—	—	μ A	Maximum allowed values: Type B or BF AP: 10 μ A NC; 100 μ A NC; Type CF AP: 10 μ A NC;
NC S5=1;S7=1;	240	50	0.26	Between reference earth and Type CF applied parts
NC S5=1;S7=0;	240	50	0.32	
NC S5=0; S7=1	240	50	0.35	
NC S5=0; S7=0	240	50	0.45	
Fig. 15 and 20 – Total Patient Leakage Current with all AP of same type connected together	—	—	μ A	Maximum allowed values: Type B or BF AP: 50 μ A NC; 500 μ A NC Type CF AP: 50 μ A NC
NC S5=1;S7=1;	220	50	1.05	In compliance, performed on all leads connected together, Type CF applied parts
NC S5=1;S7=0;	220	50	1.11	
NC S5=0; S7=1	220	50	1.14	
NC S5=0; S7=0	220	50	1.24	
Fig. 17 and 20 – Total Patient Leakage Current with all AP of same type connected together with external voltage on SIP/SOP	—	—	μ A	Maximum allowed values: Type B or BF AP: 50 μ A NC; 500 μ A NC; Type CF AP: 50 μ A NC; 100 μ A
NC S5=1;S7=1;	240	50	1.05	In compliance
NC S5=1;S7=0;	240	50	1.11	
NC S5=0; S7=1	240	50	1.14	
NC S5=0; S7=0	240	50	1.24	
Fig. 16 and 20 – Total Patient Leakage Current with all AP of same type connected together with external voltage on F-type AP	—	—	μ A	Maximum allowed values: Type B: NA Type BF: 5000 μ A Type CF: 100 μ A
				Same results as above
Fig. 18 and 20 – Total Patient Leakage Current with all AP of same type connected together with external voltage on metal Accessible Part not Protectively Earthed	—	—	—	Maximum allowed values: Type B & BF: 1000 μ A Type CF: N/A

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				N/A, all metal accessible parts are protectively earthed
Function Earth Conductor Leakage Current (FECLC)	—	—	—	Maximum allowed values: 5 mA NC; 10 mA SFC
				No functional earth

Supplementary information:

Note 1: For EARTH LEAKAGE CURRENT see 8.7.3 d) and 8.7.4.5;

Note 2: For TOUCH CURRENT see 8.7.3 c) and 8.7.4.6;

Note 3: For PATIENT LEAKAGE CURRENT SEE 8.7.3.b) and 8.7.4.7

Note 4: Total PATIENT LEAKAGE CURRENT values are only relative to equipment with multiple APPLIED PARTS of the same type. See 8.7.4.7 h). The individual APPLIED PARTS complied with the PATIENT LEAKAGE CURRENT values.

Note 5: In addition to conditions indicated in the Table, tests conducted at operating temperature and after humidity preconditioning of 5.7, EQUIPMENT energized in stand-by condition and fully operating, max rated supply frequency, at 110 % of the max RATED MAINS VOLTAGE, and after relevant tests of Clause 11.6 (i.e., overflow, spillage, leakage, ingress of water and particulate matter, cleaning & disinfection, & sterilization).

ER - Earth leakage current

TC – Touch current

P - Patient leakage current

PA – Patient auxiliary current

TP – Total Patient current

PM - Patient leakage current with mains on the applied parts

MD - Measuring device

1 - Switch closed

0 - Switch open

S5- Switch for reverse polarity

S7- Protective earth switch

NC - Normal condition

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Attachment-2 Images of Hardware



***** END OF THE TEST REPORT*****